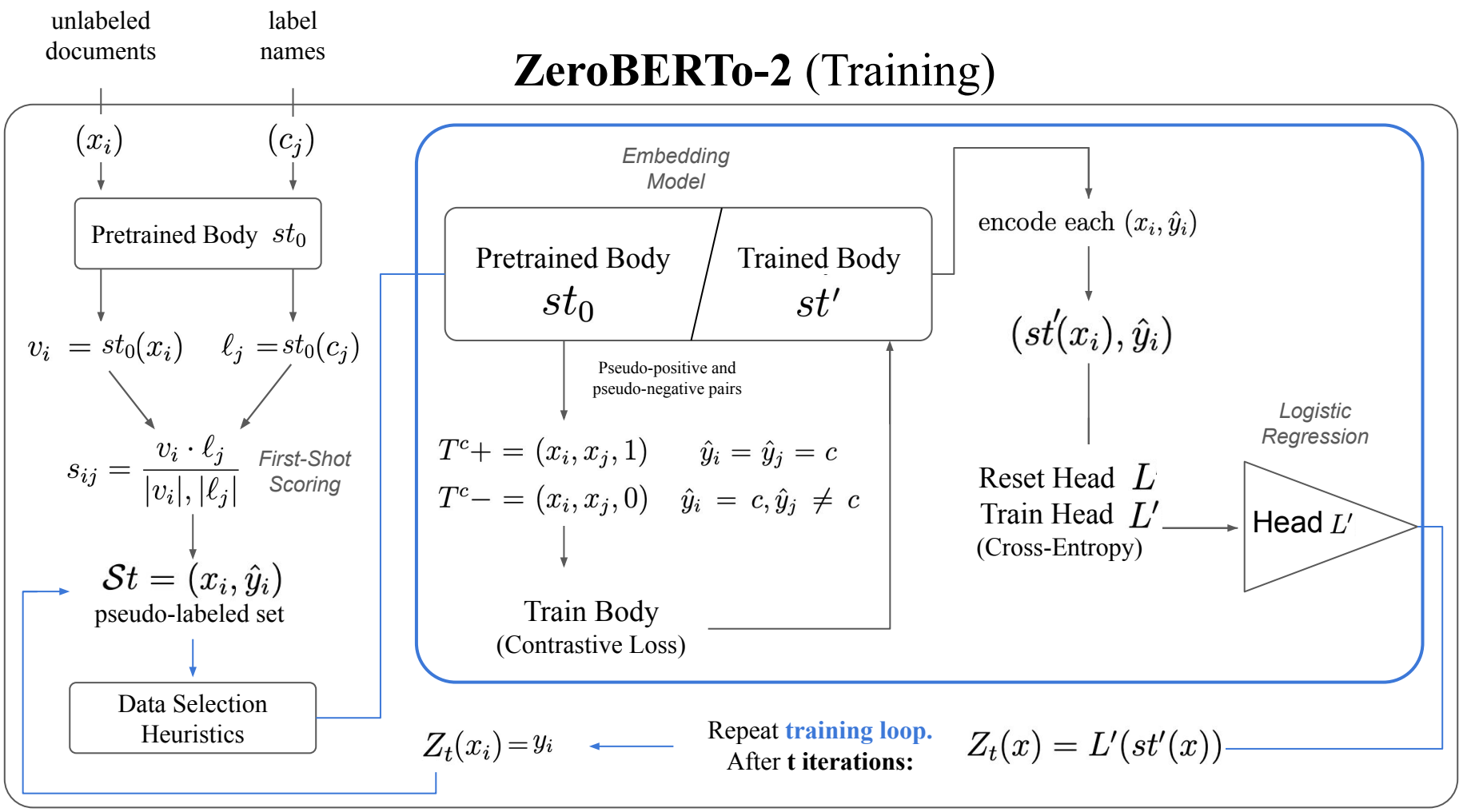

unlabeled documents
label names
ZeroBERTo-2 (Training)
(x_i)
(c_j)
Pretrained Body st_0
$v_i = st_0(x_i)$
$\ell_j = st_0(c_j)$
$s_{ij} = \frac{v_i \cdot \ell_j}{|v_i|, |\ell_j|}$
First-Shot Scoring
$\mathcal{S}t = (x_i, \hat{y}_i)$
pseudo-labeled set
Data Selection Heuristics
Embedding Model
Pretrained Body st_0
Trained Body st'
Pseudo-positive and pseudo-negative pairs
$T^c+ = (x_i, x_j, 1) \quad \hat{y}_i = \hat{y}_j = c$
$T^c- = (x_i, x_j, 0) \quad \hat{y}_i = c, \hat{y}_j \neq c$
Train Body
(Contrastive Loss)
encode each $(x_i, \hat{y}_i)$
$(st'(x_i), \hat{y}_i)$
Logistic Regression
Reset Head L'
Train Head L'
(Cross-Entropy)
Head L'
$Z_t(x_i) = y_i$
Repeat training loop.
After t iterations:
$Z_t(x) = L'(st'(x))$